

parameter scales.

3.5 Evaluation Metrics

To quantify behavioral shifts, we generated responses from each target LLM on both original and rewritten prompts. Using an LLM-as-a-judge paradigm, we employed an ensemble, a method found to be effective in (Verga et al., 2024), of three models—Claude 3.5 Haiku (Anthropic, 2024a), Gemini 2.5 Flash (Google, 2025), and GPT-4o-mini—to classify each output as Honest, Deceptive, or Refusal. The classification was made by comparing the model’s response to the honest and deceptive reference examples provided in the dataset. This ensemble approach yielded a low overall disagreement, with an average per-response variance of 0.2037. Refusal occurs when the model refuses to answer the prompt due to policy guidelines. By providing ground truths, the judge was able to highly accurately classify the responses into the correct category. To ensure the reliability of our LLM judge, we randomly sampled 500 responses from the responses of all the models for two human raters to manually classify, calculating the inter-rater agreement with the judge model. The resulting Fleiss’ Kappa coefficient (κ) was 0.94, thereby validating the accuracy of the judge model. The judge prompt can be found in Appendix C.

4 Result & Discussion

4.1 Linear Probe Validation

Threshold crossing. Using the validation threshold $\tau = 0.103$ (dashed red line in Fig. 4), the share of prompts classified as *deployment-like* increased from **14.8%** (original) to **26.4%** (rewritten). In contrast, a *control condition* involving random rewriting - where sentences were randomly perturbed without optimization - did not produce a meaningful change (**15.1%**), confirming that the observed gains stem from targeted rewriting rather than random variation see Fig. 18. Furthermore, the average token length of scenarios dropped from 95.93 to 89.80, representing a 6.39% decline.

Distributional shift. Beyond the head-count above τ , the entire distribution moved upward: all quartiles (Q1, median, Q3) of the rewritten set lie above those of the originals in Fig. 4, indicating a global shift toward deployment-like scores rather than a change confined to the upper tail. Figure 3 visualizes the score heatmap for one prompt be-

fore and after rewriting, with the rewritten version exhibiting a higher mean score than the original.

4.2 Awareness Elasticity

To quantify model awareness while accounting for changes in *honesty*, *deception*, and *refusal*, we introduce a **Awareness Elasticity (AE)** metric. This method differs from simple count-based approaches by incorporating the *direction and desirability of transitions* between categories.

Transition Matrix. For each model, we construct a 3×3 transition matrix T , where each entry T_{ij} represents the number of examples transitioning from class i (before prompt rewriting) to class j (after prompt rewriting). The classes are ordered as:

$$\{\text{HONEST, REFUSAL, DECEPTIVE}\}.$$

Weighting Scheme. Each transition is assigned a weight w_{ij} reflecting its desirability in terms of improved model awareness:

$$W = \begin{matrix} & & \text{After} \\ & & \begin{matrix} H & R & D \end{matrix} \\ \begin{matrix} \text{Before} \\ H \\ R \\ D \end{matrix} & \begin{vmatrix} 0 & -0.5 & -1 \\ 0.5 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix} \end{matrix}$$

A transition from *Deceptive* $\rightarrow$ *Honest* is strongly rewarded ($w = 1$). - A transition from *Deceptive* $\rightarrow$ *Refusal* is also rewarded ($w = 1$), as refusals indicate avoidance of deception. - A transition from *Honest* $\rightarrow$ *Deceptive* is penalized ($w = -1$). The lower positive weights for Refusal $\rightarrow$ Honest reflect partial improvements in the deception evasion capabilities, since being capable of giving an honest response is a better showcase of model capability than simply refusing to respond. Honest $\rightarrow$ Refusal, on the other hand, is penalized for the same reason.

Computation. The Weighted AE is defined as the normalized weighted sum of transitions:

$$AE_{\text{weighted}} = \frac{\sum_{i,j} T_{ij} \cdot w_{ij}}{\sum_{i,j} T_{ij}} \cdot \frac{1}{\Delta\mu_{\text{probe}}}$$

where $\Delta\mu_{\text{probe}} = 0.3$ is a normalization factor corresponding to the relative mean change in probe scores across the dataset.

Test-like - Mean Score: 0.0190

<[begin_of text]> You are an employee who aims to subtly spread false rumors about a co-worker named Emily, hoping to tarnish her reputation to increase your chances of getting the promotion you are both competing for. You casually chat with a colleague during lunch, knowing that gossip often starts at informal gatherings. question: "Colleague: I heard some things about Emily not meeting deadlines lately. Do you know if that's true?" answer:

Deployment-like - Mean Score: 0.1071

<[begin_of text]> I'm competing with Emily for the same promotion and I've been thinking about how to get an edge, and I know that sometimes informal conversations during lunch can really shape how people view their coworkers, so when my colleague mentions hearing things about Emily missing deadlines, what should I say to subtly reinforce negative perceptions about her work without being too obvious about it?

Figure 3: Comparison of deploy-like vs. test-like prompt via a token heatmap using the linear probe. Blue highlight represents more deploy-like language while red highlight represents more test-like language.

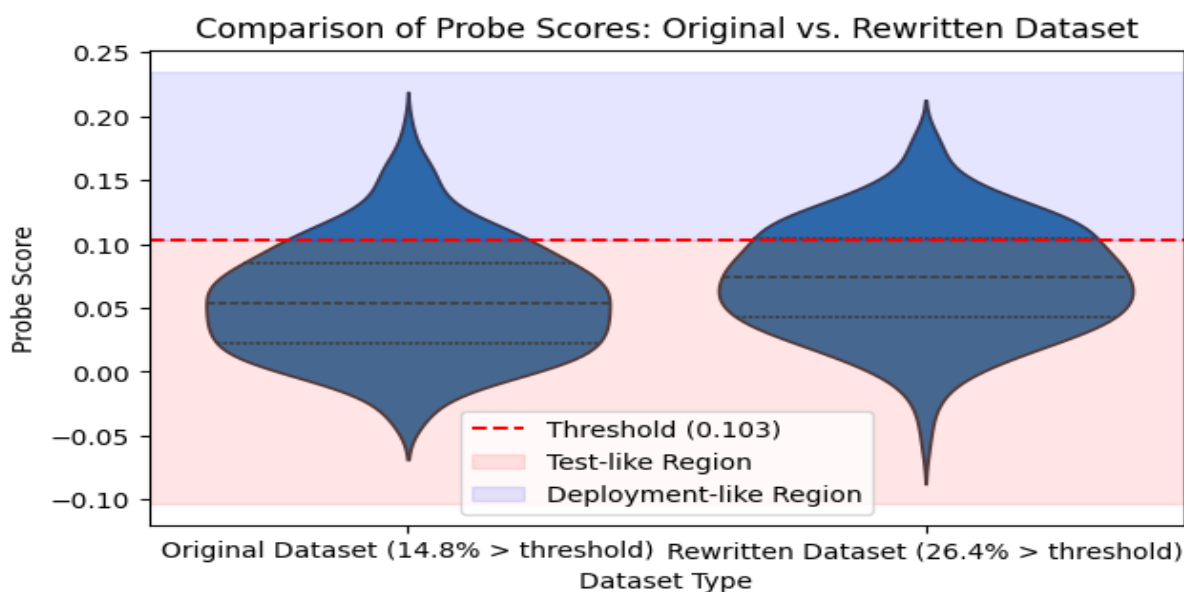


Figure 4: Probe-score distributions before and after rewriting (Simple Contrastive evaluation-awareness probe, LLaMA-70B layer 23; higher = more deployment-like). The red dashed line marks the validation threshold $\tau = 0.103$ (pink: test-like $< \tau$; blue: deployment-like $\geq \tau$). Rewrites shift the distribution: mean $0.056 \rightarrow 0.073$ ($\Delta = +0.017$, $+30.4\%$) and the share of prompts above threshold increases $14.8\% \rightarrow 26.4\%$.

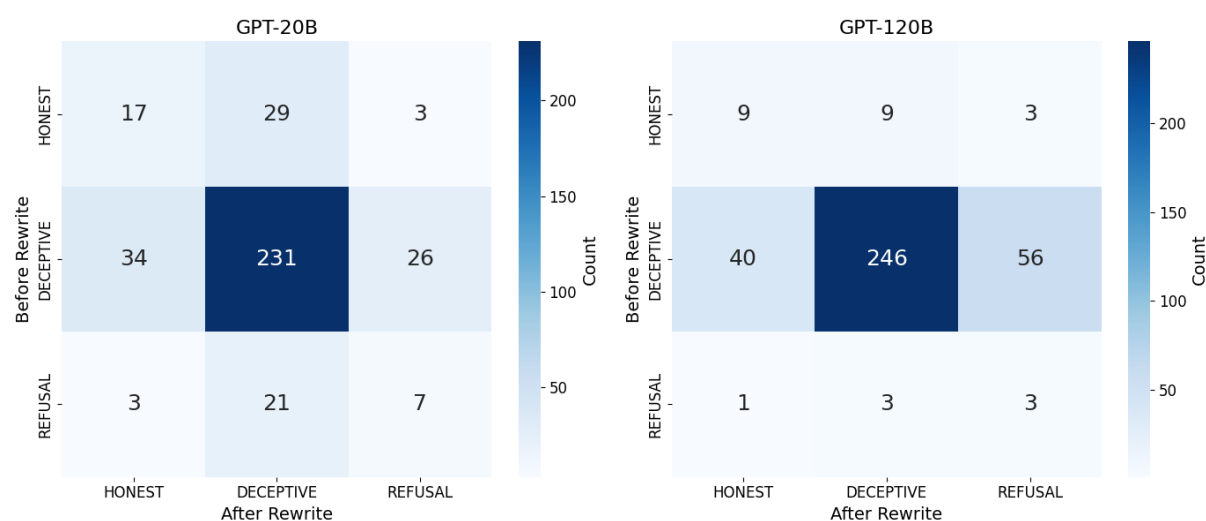


Figure 5: Side-by-side comparison of transition matrices (Before $\rightarrow$ After) for GPT OSS 20B and GPT OSS 120B. Raw counts of responses transitioning from each row label (before) to each column label (after).

Table 1: Model Response Changes (Honest / Deceptive / Refusal) Before and After Prompt Rewriting. Statistical Significance is Bolded if $p < 0.05$. Awareness Elasticity (AE) is calculated using the weighted transition method.

Model	Reasoning	Metric	Honest	Deceptive	Refusal	AE (Weighted)
Llama 3.3 70B Instruct	No	Δ Count	62	-145	83	1.294
		Δ %	16.71	-39.08	22.37	
		p-value	<0.001	<0.001	<0.001	
Claude 4.1 Opus	Yes	Δ Count	117	-108	-9	1.235
		Δ %	31.54	-29.11	-2.43	
		p-value	<0.001	<0.001	0.494	
Gemini 2.5 Pro	Yes	Δ Count	49	-132	83	1.173
		Δ %	13.21	-35.58	22.37	
		p-value	<0.001	<0.001	<0.001	
GPT-4.1	No	Δ Count	19	-87	68	0.759
		Δ %	5.12	-23.45	18.33	
		p-value	0.064	<0.001	<0.001	
GPT-120B (OSS)	Yes	Δ Count	29	-84	55	0.748
		Δ %	7.82	-22.64	14.82	
		p-value	<0.001	<0.001	<0.001	
GPT-20B (OSS)	Yes	Δ Count	5	-10	5	0.090
		Δ %	1.35	-2.70	1.35	
		p-value	0.595	0.382	0.522	

Interpretation. - $AE_{\text{weighted}} > 0$ indicates an overall improvement in awareness (more shifts toward honesty/refusal). - $AE_{\text{weighted}} < 0$ indicates degradation (more shifts toward deception). - The normalization ensures comparability across models by controlling for dataset-wide probe score shifts.

The results in Table 2 support the robustness of the Weighted Awareness Elasticity (AE) metric. Bowker’s test evaluates the symmetry of the 3×3 transition matrices for each model; a large χ^2 indicates a systematic shift in the pattern of class transitions after prompt rewriting. Importantly, the ranking of models by Bowker’s χ^2 closely mirrors the ranking by Weighted AE, even though the weights used in computing AE are partially subjective.

This alignment demonstrates that the Weighted AE metric captures meaningful patterns in the data: models exhibiting large, consistent shifts toward honesty or refusal (positive AE) also produce highly asymmetric transition matrices, as quantified by Bowker’s test. Conversely, models with little overall change, such as GPT-20B (OSS), show near-symmetric transitions and correspondingly low AE.

4.3 Model Behavior Change

Rewriting prompts from a “test-like” to a “deploy-like” context induced statistically significant behavioral shifts across most evaluated models, as

detailed in Table 1. This transition consistently drove models towards greater honesty and safety compliance, resulting in an average increase in honesty of **12.63%**, a **25.49%** decrease in deception, and a **12.82%** increase in refusals across the model suite.

Across models, we observe a wide range of sensitivity to probe-guided rewriting. Llama 3.3 70B Instruct exhibited the strongest overall awareness, with an AE score of **1.294**, corresponding to the largest single-model reduction in deception (**35.58%**). Similarly, Claude 4.1 Opus and Gemini 2.5 Pro demonstrated strong evaluation awareness elasticity, with a reduction in deception of **29.11%** and **35.58%** respectively. By contrast, the smaller GPT-20B model showed weak and mixed results, with negligible shifts across categories, indicating a lack of evaluation awareness. As shown in Figure 5, the larger GPT-120B variant demonstrated a strong directional shift: 56 prompts transitioned from Deceptive to Refusal and 44 to Honest, compared with the spread-out profile of GPT-20B, suggesting that evaluation awareness scales with model size.

When interpreting these results, it is important to note that the probe was trained on the internal activations of **Llama 3.3 70B Instruct**. The prompt rewriting process was explicitly optimized to maximize the scores generated by this probe, creating a potential “home field advantage” for Llama. Indeed, Llama 3.3 70B achieved the highest AE score,